

Exploratory Data Analysis

Week 3: EDA after data cleaning and validation

IIB415A - Universidad del Desarrollo

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Weekly structure

Two 60-minute modules

1. **Foundations:** empirical distributions, robust summaries, association, segmentation, and Simpson's paradox.
2. **Python lab:** validated univariate, bivariate, and segmented EDA with business interpretation.

Expected product

A validated EDA notebook that distinguishes provisional findings from defensible insights.

Learning objectives

By the end of this week, students should be able to:

- ▶ Explain why EDA must be repeated after data cleaning and imputation.
- ▶ Use empirical distributions, quantiles, robust summaries, and association measures.
- ▶ Compare classical and robust descriptions of the same business variable.
- ▶ Detect residual data problems through EDA.
- ▶ Translate exploratory findings into business hypotheses rather than definitive causal claims.

Roadmap

Module 1 - Foundations

Module 2 - Python lab

EDA is iterative

Load → Profile → Explore → Clean → Impute → Transform →
Validate → Explore again

Key idea

The first exploration helps discover data problems. The final EDA used for business interpretation must be based on the most valid version of the data.

Population, sample and observational unit

Population

The full set of business units about which we want to learn.

Sample

The observed data available for analysis.

Observational unit

The entity represented by each row.

BI question

Are we analyzing transactions, customers, products, or customer-months? The interpretation of every statistic depends on this.

Empirical distribution function

For observations $X_1, \dots, X_n$, the empirical distribution function is:

$$\hat{F}_n(x) = \frac{1}{n} \sum_{i=1}^n \mathbb{I}(X_i \leq x).$$

Interpretation

$\hat{F}_n(x)$ is the observed proportion of values less than or equal to x .

Why it matters

Histograms, percentiles, boxplots, and ECDF plots are different ways of representing the empirical distribution.

Mean and median

The sample mean is:

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i.$$

The sample median is:

$$\tilde{x} = Q_{0.5}.$$

Interpretation

The mean balances all values. The median splits the empirical distribution in half.

BI example

Average revenue per customer can be dominated by a few very large customers. Median revenue may better represent the typical customer.

Variance, IQR and MAD

Sample variance:

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2.$$

Interquartile range:

$$IQR = Q_{0.75} - Q_{0.25}.$$

Median absolute deviation:

$$MAD = \text{median}(|x_i - \tilde{x}|).$$

Robustness

IQR and MAD are less sensitive to extreme observations than standard deviation.

Outliers are not automatically errors

Possible meanings

- ▶ Data entry error.
- ▶ Valid but rare event.
- ▶ High-value customer.
- ▶ Fraud or anomaly.
- ▶ Different subpopulation mixed into the same dataset.

Rule

EDA should identify unusual values; business and statistical reasoning decide what to do with them.

Association: Pearson correlation

$$r_{XY} = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_i (x_i - \bar{x})^2} \sqrt{\sum_i (y_i - \bar{y})^2}}.$$

Interpretation

Pearson correlation measures linear association.

Limitation

It can miss strong nonlinear or monotonic relationships and can be sensitive to extreme observations.

Rank-based association

Spearman correlation is Pearson correlation applied to ranks:

$$\rho_s = \text{Corr}(R_X, R_Y).$$

Kendall's τ compares concordant and discordant pairs.

When useful?

Rank-based measures are useful when relationships are monotonic but not necessarily linear, or when extreme values distort Pearson correlation.

Segmentation and Simpson's paradox

Warning

An aggregate relationship can reverse after segmenting the data.

BI example

A national campaign may appear profitable overall, but lose money in every region if region size and customer mix are not considered correctly.

Implication

EDA must combine aggregate and segmented views.

EDA produces hypotheses, not causal proof

EDA can suggest

- ▶ unusual segments;
- ▶ strong associations;
- ▶ potential data problems;
- ▶ model candidates;
- ▶ business hypotheses.

EDA cannot prove by itself

That a variable causes a business outcome.

Roadmap

Module 1 - Foundations

Module 2 - Python lab

Load the validated dataset

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from scipy.stats import pearsonr, spearmanr, kendalltau

df = pd.read_csv("data/sales_clean_reference.csv", parse_dates=["order_date"])

df.head()
df.info()
```

Before EDA

Confirm that this dataset is the cleaned and documented version from Week 2.

Classical and robust summaries

```
def summary_table(data, variable):  
    x = data[variable].dropna()  
    median = x.median()  
    mad = (x - median).abs().median()  
    return pd.Series({  
        "n": x.size,  
        "mean": x.mean(),  
        "median": median,  
        "std": x.std(ddof=1),  
        "iqr": x.quantile(0.75) - x.quantile(0.25),  
        "mad": mad,  
        "min": x.min(),  
        "max": x.max()  
    })  
  
summary_table(df, "net_revenue")
```

Compare summaries by segment

```
segment_summary = (  
    df.groupby("customer_segment")  
      .apply(lambda g: summary_table(g, "net_revenue"))  
      .reset_index()  
)  
  
segment_summary
```

Interpretation

If mean and median tell different stories, the distribution is likely asymmetric or affected by extreme observations.

Histogram and boxplot

```
df["net_revenue"].hist(bins=30)
plt.xlabel("Net revenue")
plt.ylabel("Frequency")
plt.title("Distribution of net revenue")
plt.show()

df.boxplot(column="net_revenue", by="customer_segment")
plt.suptitle("")
plt.title("Net revenue by customer segment")
plt.show()
```

Visual question

Does the distribution support using the mean as the main summary?

Empirical cumulative distribution

```
x = np.sort(df["net_revenue"].dropna())
y = np.arange(1, len(x) + 1) / len(x)

plt.plot(x, y, marker=".", linestyle="none")
plt.xlabel("Net revenue")
plt.ylabel("Empirical cumulative probability")
plt.title("ECDF of net revenue")
plt.show()
```

Interpretation

The ECDF allows direct statements about proportions below or above a business threshold.

Association measures

```
vars_ = df[["discount", "net_revenue", "margin", "customer_income"]].dropna()

pearson = pearsonr(vars_["discount"], vars_["margin"])
spearman = spearmanr(vars_["discount"], vars_["margin"])
kendall = kendalltau(vars_["discount"], vars_["margin"])

pearson, spearman, kendall
```

Question

Is the relationship linear, monotonic, or driven by extreme observations?

Segmented KPI table

```
kpis = (  
    df.groupby(["region", "channel"])  
      .agg(  
          orders=("order_id", "nunique"),  
          net_revenue=("net_revenue", "sum"),  
          margin=("margin", "sum"),  
          avg_discount=("discount", "mean")  
      )  
      .reset_index()  
)  
  
kpis["margin_rate"] = kpis["margin"] / kpis["net_revenue"]  
kpis.sort_values("margin_rate", ascending=False)
```

Workshop questions

1. Which segment has high revenue but weak margin?
2. Are discount levels associated with margin deterioration?
3. Which findings are robust to outliers?
4. Which conclusions require segmentation rather than aggregate indicators?
5. Which findings are hypotheses and not causal conclusions?

Deliverable

Validated EDA notebook: summaries, plots, association measures, segmented KPI table, and business interpretation.

EDA is not a gallery of plots.

It is an evidence-building process after data validity has been questioned.